

Charotar University of Science and Technology  
M.Sc (Advance Organic Chemistry) Semester- II Examination May 2011  
OC -711: Inorganic Chemistry

Date: May 06, 2011 Time: 10.00 am to 10.30 am

Max. Marks: 20

Part A

1. For  $d^2$  configuration the ground state is : (2)  
i.  $^4F_{3/2}$  ii.  $^3F_2$  iii.  $^5D_0$  iv.  $^2D_{5/2}$
2. In square pyramidal transition metal complex the highest energy level is: (2)  
i.  $d_{x^2-y^2}$  ii.  $d_{z^2}$  iii.  $d_{xy}$  iv.  $d_{xz}$
3. For Cu(II) in  $O_h$  field the electronic transition is: (2)  
i.  $E_g \rightarrow T_{2g}$  ii.  $T_{2g} \rightarrow E_g$  iii.  $T_{1g} \rightarrow T_{2g}$
4. The number of microstates for  $d^3$  configuration is: (3)  
i. 20, ii. 120, iii. 45, iv. 91
5. Neel temperature is characteristic of : (2)  
i. anti ferromagnetic compound, ii. ferromagnetic compound  
iii. paramagnetic compound.
6. The most unstable orbital in trigonal bipyramidal geometry is : (2)  
i.  $d_z^2$  ii.  $d_{x^2-y^2}$  iii.  $d_{xz}$  iv.  $d_{yz}$
7.  $[\text{Ni}(\text{CN})_4]^{2-}$  is diamagnetic. Explain with the help of valence bond theory. (3)

8. Explain, whether orbital contribution to magnetic moment is expected or not in:

(2)

i. High spin octahedral  $d^6$  complex.

ii. High spin octahedral  $d^3$  complex.

iii. Low spin tetrahedral  $d^4$  complex.

iv. High spin tetrahedral  $d^4$  complex.

9. Construct the  $\sigma$  LGO having  $A_{1g}$  symmetry for an octahedral complex.

(2)



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Part B

- 1(a). Predict whether a high spin or a low spin complex will be favored for  $Mn^{+3}$  in  $O_h$  when the ligand has 10 Dq value of  $21000\text{ cm}^{-1}$ . The pairing energy is  $28000\text{ cm}^{-1}$ . (4)
- (b). In oxovanadium (iv) complex, the relative position of  $d_{z^2}$  and  $d_{x^2-y^2}$  gets interchanged as compared to square pyramidal. Explain. (3)
- (c). Explain the splitting of d orbitals either in a tetrahedral field or in a trigonal prismatic stereochemistry. (3)
- 2(a). Write in brief about Jahn –Teller effect in transition metal complexes. (3)
- (b). Draw the Orgel diagram for  $d^n$  (where  $n=2,3,7,8$ ) metal ions in octahedral and tetrahedral crystal fields. (3)
- (c). Three spin allowed transitions are observed at  $17000\text{ cm}^{-1}$ ,  $25000\text{ cm}^{-1}$  and  $36000\text{ cm}^{-1}$  for a V(III) complex. Calculate the values of 10 Dq, B,  $\beta$  and  $\beta\%$ , given  $B_0=860\text{ cm}^{-1}$ . (4)
- OR
- (c). Obtain the values of 10Dq, B,  $\beta$  and  $\beta\%$  for  $[Co(H_2O)_6]^{+2}$ , given that  $B_0=971\text{ cm}^{-1}$  and there are three spin allowed transitions at  $8100\text{ cm}^{-1}$ ,  $16000\text{ cm}^{-1}$  and  $19500\text{ cm}^{-1}$ .
- 3(a). The  $d_{z^2}$  orbital is the most unstable in trigonal bipyramidal geometry. Justify. (3)
- (b). Demonstrate, how the RS states of  $d^2$  configuration are obtained. (3)
- (c). The complex  $[Cr(H_2O)_6]Cl_3$  records three spin allowed transitions at  $17400\text{ cm}^{-1}$ ,  $24600\text{ cm}^{-1}$  and  $37800\text{ cm}^{-1}$ . Obtain the values of 10Dq, B,  $\beta$  and  $\beta\%$ , given that  $B_0=918\text{ cm}^{-1}$ . Also, calculate the energy of interaction between the two  $^4T_{1g}$  states. (4)
- 4(a). Derive Van Vleck's equation for magnetic susceptibility. (6)
- (b). What is spectrochemical series? Explain the magnetic behavior of octahedral  $[Fe(CN)_6]^{4-}$  and  $[Fe(H_2O)_6]^{2+}$  in terms of crystal theory. (4)
- 5(a) With the help of molecular orbital diagram explain the magnetic behavior of  $[CoF_6]^{3-}$  (6)
- (b). A Cr(III) ion in an octahedral field has  $^4A_{2g}$  ground state. Derive an expression for the temperature dependence of the magnetic susceptibility for  $[CrL_6]^{+3}$  complexes, L is a (4)

neutral ligand.

OR

- (b). Discuss in brief the super exchange mechanism of anti-ferromagnetism. Which one of the two compounds,  $[(\text{NH}_3)_5\text{Cr} - \text{O} - \text{Cr}(\text{NH}_3)_5]\text{Br}_4$  (have  $\text{Cr}-\text{O}-\text{Cr}$  linear) and  $[(\text{NH}_3)_5\text{Cr} - \text{OH} - \text{Cr}(\text{NH}_3)_5]\text{Br}_5$  (have  $\angle \text{O}-\text{Cr}-\text{O}$  as  $\sim 120^\circ$ ) will have more effective super exchange interaction?